Clark University

Running a Non-Profit in a Recession: Analysis of Countercyclical Financial Practices

Economics Honors Thesis

by

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2026

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Introduction

Contrary to publicly traded for-profit companies, non-profits do not need to seek continual expansion; rather, they respond to changing demands and needs of the community they volunteer to support. Theoretically then, many non-profits do not need to expand when the broader economy is performing well – because their constituents will be better off and in less need for philanthropy. However, in an economic downturn or recession, demand for nonprofit services inherently increases, suggesting that nonprofits should counter-cyclically expand their program offerings to match this heightened need. Despite this expectation of altruistic countercyclicality, many non-profits are constrained by their budget and available revenues, which follow a procyclical cycle. In a recession, investment income crashes, states shrink their budgets, and individual giving falls. But not all revenue sources for a given organization fall in a recession – some organizations may save surpluses from good for bad times, the federal government may pass a non-profit stimulus package, and organizations can reshuffle assets.

This creates a puzzle: what financial practices are most correlated with counter cyclicality in US 501(c)(3) non-profit organizations?

To answer this, I conduct a longitudinal panel data regression analysis utilizing digital Form-990 tax returns from tax-exempt organizations across the United States. The final dataset comprises nearly 2.88 million observations from 391,529 unique organizations spanning a 15-year period from 2009 to 2023. Since non-profits lack a simple "bottom line," I use program expenses as a proxy for the demand-side output of mission fulfillment, and total revenues to represent the supply-side input of peak capacity. I use an Ordinary Least Squares (OLS) model with year, state, industry, and firm fixed effects to observe the correlations between financial behaviors and economic conditions.

The findings confirm that the non-profit sector is largely procyclical, aligning with recent literature. The firm fixed-effects model reveals that for a one percentage point increase in real state GDP, a non-profit's total revenue and program expenses both increase by approximately 0.6%. However, when controlling for a budget constraint, the economic conditions of a non-profit's state no longer have a significant effect. Instead, the change in program expenses is mostly explained by changes in total revenue. The revenue-controlled model reveals for a one percent increase in a non-profit's total revenue, their program expenses increase by .82%. However, countercyclical behavior is still observable across some industries of organizations and different revenue sources. Furthermore, when testing against the specific macroeconomic shock of the Covid-19 recession, the models reveal that countercyclical behavior probability is significantly correlated on an organization's revenue streams.

While existing literature has confirmed the procyclical trend of larger non-profit enterprise and the increased demand by constituents as the economy declines, these studies have historically been limited by a narrow focus on individual recessions. Literature investigating optimal non-profit practices is limited to case studies of specific industries or revenue models. This thesis contributes to the existing body of literature by observing multiple industries, geographies, and organizations across a large timeframe to find resilient practices across the entire non-profit enterprise in the United States.

Literature Review

Research into non-profit economics was limited for a long time but has been continually growing in volume in the past few decades, moving from abstract theory to empirical study of financial practices. Hansmann (1980) wrote some of the earliest and influential theoretical work on the role of nonprofits, establishing the "nondistribution constraint" that distinguishes non-profits from for-profit firms. Although not technically barred from making a profit, a non-profit corporation can only use the surplus for reinvesting in capital to advance the non-profit's stated mission. This difference distinguishes non-profit operations because it should vastly improve trust among consumers and encourage the expansion of nonprofit enterprises across various sectors of the economy.[1] Although more concerned with legal perspective than economic and theoretical, Hansmann research is foundational and expanded upon, including in economics. Hansmann's theoretical argument can establish why non-profits may operate outside of typical, cyclical trends and require unique financial procedures.

The transition toward a formal economic perspective of non-profit research accelerated after work from Rose-Ackermann (1996), who directly confronted economists' assumptions that non-profits were "noneconomic institutions."[2] Using empirical economic and psychological methods, Rose-Ackermann supports economic logics for altruism, nonprofits' role in solving information asymmetries, and the high trust perceived in non-profit corporations. Her research is important because it was some of the earliest encouragement for using regression analysis to analyze non-profit behavior over time. However, while Rose-Ackermann provided the early map for empirical study, her work lacked large-scale panel data to see how organizations behave over

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[1] Hansmann. "The Role of Nonprofit Enterprise."
[2] Rose-Ackerman, Susan. "Altruism, nonprofits, and economic theory." Journal of economic literature 34, no. 2 (1996): 701-728.

time. My study fills this gap by applying these economic logics to a longitudinal dataset of almost 2.9 million observations.

Despite the theoretical potential for non-profits to be countercyclical, the work by Exley, Lehr, and Terry (2023) quantified a problematic procyclical trend by analyzing the Covid-19 recession. Their research found that during an economic downturn, non-profit organizations commonly shrink their financial size. Revenues, expenses, and balance sheets are all strongly procyclical, disappointing public expectations for certain industries to expand countercyclically, such as those providing housing and food assistance.[3] Similarly, researchers Bridgeland, McNaught, Reed, and Dunkelman (2010) focused on what they termed the "quiet crisis" of shrinking non-profit sector following the 2008 recession. They find equivalent results: organizations faced diminishing charitable donations, reduced state and local government budgets, and increasing human need, which resulted in large non-profit budget cuts, layoffs, and halted programs, especially across human service and cultural organization industries.[4] While these studies identify the problem during specific macroeconomic shocks, their shortcoming is this narrow focus on an individual recession. My research expands on their work by looking at a 14-year period (2009–2023) to see if this procyclicality is an inherent trait of non-profit enterprise or just an observable reaction to a recession.

Building on the observation that nonprofits are highly vulnerable to economic downturns, Arbogust (2020) compounds the results of Exley et al[5] and Bridgeland et al. [6] by conducting a longitudinal analysis of specific financial practices over time that can predict future failure of

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[3] Exley, Christine L., Nils H. Lehr, and Stephen J. Terry. "Nonprofits in good times and bad times." Journal of Political Economy Microeconomics 1, no. 1 (2023): 42-79.
[4] Bridgeland, John M., Mary McNaught, Bruce Reed, and Marc Dunkelman. "The quiet crisis: The impact of the economic downturn on the nonprofit sector." Civic Enterprises (2009).
[5] Exley, Lehr, and Terry. "Nonprofits in good times and bad times."
[6] Bridgeland et al. "The quiet crisis"

non-profit organizations. Significant changes in revenue concentration, administrative costs, or new unrelated business income are the main statistical differences between the organizations that fail versus ones that continue to succeed.[7] The vulnerabilities identified in these studies motivate my research into financial practices that improve countercyclical resilience. I seek to use a similar method, but much simpler model, as Arbogust to find practices correlated with 'success' of a non-profit.

Recent research into optimal financial practices has begun to challenge 'conventional wisdom' by identifying how specific management choices can improve resilience. For example, Altamimi and Liu's (2021) research into the optimal overhead ratio challenges the assumption to 'keep it as low as possible' by demonstrating an upside-down U-shaped relationship where increased overhead spending improves program outcomes, up to an optimal point.[8] Further, Jones, Kitching, Roberts, and Smith's (2012) research about how well-managed non-profits can 'weather the storm' by saving revenues demonstrates that charities can actively smooth their spending over good and bad economic times by saving a substantial portion of revenue increases and then intentionally using those savings to mitigate future revenue reductions when the economy worsens.[9] However, this type of research is traditionally limited to a 'case study' or specific group of organizations (i.e. arts/culture non-profit industry or non-profits that rely on independent donations as their revenue source) and so results cannot be confidently generalized to firms outside of this scope. I hope to address this shortcoming by analyzing multiple

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[7] Arbogust, MacKenzie. Why do nonprofits fail? A quantitative study of form 990 information in the years preceding closure. Old Dominion University, 2020.
[8] Altamimi, Hala, and Qiaozhen Liu. "The nonprofit starvation cycle: Does overhead spending really impact program outcomes?." Nonprofit and Voluntary Sector Quarterly 51, no. 6 (2022): 1324-1348.
[9] Jones, Christopher L., Karen A. Kitching, Andrea Alston Roberts, and Pamela C. Smith. "The spend-save decision: An analysis of how charities respond to revenue changes." Accounting Horizons 27, no. 1 (2013): 75-89.

industries, geographies, and types of organizations to hopefully find financial practices that are consistent across these large groups.

Also following a broad scope of organizations, specifically all non-profits following the Italian economic crisis in 2008, Pollifroni, Militaru, and Socaciu (2014) establish economic theory to support non-profit countercyclicality.[10] They argue that countercyclicality arises when organizations combine the nondistribution constraint's focus on durability with a commitment to corporate social responsibility and find specific financial practices correlated with improving a non-profit's ability to be countercyclical. They build on early theoretical research and find empirical support that the "nondistribution constraint" (see Hansmann 1980[11]) of non-profits forces them to reinvest capital into the long-term durability, fulfilling the first condition. And a non-profit's voluntary goal to maximize public utility, run by unselfish stakeholders, and lack of proprietary rights fulfill the second condition for corporate social responsibility. Important to the real observable outcomes, both these conditions are necessary but not sufficient for countercyclicality. A non-profit must still actively pursue the ethical managerial style and intangible remunerations (i.e., employee benefits, remote work, and organizational flexibility) to improve countercyclicality.[12] Their study provides a strong theoretical framework and support but is limited to the Italian context, which has different national policies, culture, and business practices compared to the United States. Nonetheless, I seek a similar test of whether these same countercyclical outcomes can be observed across the diverse landscape of the United States non-profit sector.

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[10] Pollifroni, Massimo, Gheorghe Militaru and Tiberiu Socaciu. "Economic crises and the counter-cyclical profile of the Non-Profit Organisations: some empirical evidences from the Italian context." Economics, Management, and Financial Markets 9 (2014): 303-311.
[11] Hansmann. "The Role of Nonprofit Enterprise."
[12] Pollifroni, Militaru, and Socaciu. "Economic crises and the counter-cyclical profile of the Non-Profit Organisations."

Effectively measuring these trends for my thesis requires a well-supported methodology, because non-profits lack a simple "bottom line" like for-profit businesses. Unlike for-profit corporations, where research can compare net incomes to measure demand, it is impossible to choose one financial variable to generalize all non-profit goals. There is no consensus of which financial variable to use to measure non-profit demand.[13] However, Kim (2017) provides a strong justification for using program expenses as a proxy for effectiveness, finding that spending is closely correlated with actual program outcomes like attendance or registrations.[14] Furthering this, research from Ma and Beaton (2024) shows that non-profit effectiveness is tied to the stakeholders' demand for services by the organization. Specifically, organizations that are highly effective in achieving its mission at a baseline level are also able to adapt to changing demands, such as during an environmental shock like Covid-19.[15] This bridges the gap for my methodology: rather than strictly measuring consumer demand, program expenses function as a viable proxy for service provision and mission output, reflecting the organization's active effort to meet community needs.

Finally, to control for the unobservable traits of organizations across the country, I follow the recommendations by Ba, Berrett, and Coupet, who describe a high-level approach to conducting panel data research for non-profit studies. They encourage using a fixed effect OLS regression model as its best to control for unobservable traits of organizations (e.g., leadership

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[13] Boateng, Agyenim, Raphaël K. Akamavi, and Girlie Ndoro. "Measuring performance of non-profit organisations: evidence from large charities." Business Ethics: A European Review 25, no. 1 (2016): 59-74.
[14] Kim, Mirae. "The relationship of nonprofits' financial health to program outcomes: Empirical evidence from nonprofit arts organizations." Nonprofit and Voluntary Sector Quarterly 46, no. 3 (2017): 525-548.
[15] Ma, Yinglin, and Erynn E. Beaton. 2024. "Programming Change Among Nonprofit Human Service Organizations During the COVID-19 Pandemic." Human Service Organizations: Management, Leadership & Governance 48 (3): 285–304. doi:10.1080/23303131.2023.2228854.

styles, values) and larger scale effects (e.g., market structures) that will bias results.[16] By using their recommended methodological approaches on a dataset significantly larger than previous studies, I hope to identify how non-profits respond to the economic cycle and what financial practices might be most correlated with resiliency during a recession or countercyclicality.

[16] Ba, Yuhao, Jessica Berrett, and Jason Coupet. "Panel data analysis: A guide for nonprofit studies." VOLUNTAS: International Journal of Voluntary and Nonprofit Organizations 34, no. 2 (2023): 193-208.

Data Measurement and Exploration

This thesis looks to explore cyclical trends of financial practices for non-profit organizations in the United States.

Collection

For this, the dataset is made up of digital copies of Form-990 documents, which describe financial details and business operations of filers, sent by every tax-exempt organization in the United States each year from 2009 - 2023. Only businesses electronically filling for income tax exemption under section 501(c)(3) of the Internal Revenue Code, based in any of the 50 US states, with gross receipts of greater than $200,000 in one year or net assets of greater than $500,000 are included in this dataset for this thesis. I explain further restrictions to this dataset in the subheading below.

Digitally summarized copies of yearly Form-990s are published online by the Internal Revenue Services (IRS) as .XML file formats.[17] The National Center for Charitable Statistics (NCCS), a part of the non-profit and philanthropy research department of the Urban Institute, converts this raw data into data tables organized by year and header sections of the Form 990 document.[18] I compiled this collection from the NCCS of .CSV files into a raw dataset, then added supplemental data, then imposed necessary restrictions.

This preliminary dataset of Form 990 data is in Panel form. Each organization fulfilling the IRS's requirements is observed in at least one year of data. Each organization is only observed once per year; however, not all the same entities are included in each year. This makes the panel data unbalanced. Specifically, the data reports raw variables for revenues, expenses,

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[17] Irs.gov. "Form 990 Series Downloads | Internal Revenue Service," 2019. https://www.irs.gov/charities-non-profits/form-990-series-downloads.
[18] Jesse Lecy, (2023). NCCS IRS 990 Efile Data. https://urbaninstitute.github.io/nccs/datasets/efile/

activities, and governance of each organization. This includes granular financial data for the entire scope of operations (i.e., "Grants and other monetary assistance to domestic individuals," Part IX: Statement of Functional Expenses), categorical information of the entity (i.e., "Form of Organization," Part 0: Header), and summarized finances and balance sheet (i.e., "Total Assets," Part I: Summary). For this thesis, I rely on mostly summarized financial variables from the dataset.

To supplement the Form 990 dataset, the Internal Revenue Service collects Primary Activity Codes or National Taxonomy of Exempt Entities (NTEE) classification codes, but not both, for every single organization. NTEE codes are the newest format of non-profit business classification adopted by the IRS.

While filing a Form 990, or tax exemption form, an organization could list up to three activity codes to describe up to three of their main products/services provided. Exempt organizations founded before 1995 must continue using old activity codes. The activity codes for all non-profit organization, past and present, are in the Internal Revenue Service's Business Master File.[19] Organizations founded after 1995 must use the new NTEE codes. The NTEE code for all non-profit organizations, past and present, is also in the Business Master File. This master file is divided for each US state, based on organization headquarters location. For each state, there is a .CSV file. First, I combined all 50 files then coded a matching function so that organizations filing an activity code are replaced by the function to the nearest NTEE code. The IRS provides information describing the categories for both activity and NTEE codes, as well as examples of organizations belonging to each category. Using this information, I manually matched each activity code category to its nearest related NTEE code category.

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[19] Irs.gov. "Exempt Organizations Business Master File Extract (EO BMF) | Internal Revenue Service," 2017. https://www.irs.gov/charities-non-profits/exempt-organizations-business-master-file-extract-eo-bmf.

Below is a table of all NTEE code categories, with the number of observations in my final dataset (after all restrictions):

Table 1. Distribution of Non-profit Industries

Letter	NTEE Code Industry	Example Organization	Observations in Final Dataset
A	Arts, Culture & Humanities		231,063
B	Education		305,111
C	Environment		62,281
D	Animal-Related		58,215
E	Health Care		202,248
F	Mental Health & Crisis Intervention		57,354
G	Voluntary Health Associations & Medical Disciplines		29,668
H	Medical Research		23,483
I	Crime & Legal-Related		34,342
J	Employment		34,147
K	Food, Agriculture & Nutrition		37,550
L	Housing & Shelter		156,918
M	Public Safety		62,614
N	Recreation & Sports		191,978
O	Youth Development		82,292
P	Human Services		315,502
Q	International, Foreign Affairs		52,085
R	Civil Rights, Social Action & Advocacy		28,847
S	Community Improvement & Capacity Building		257,393
T	Philanthropy, Voluntarism & Grantmaking		134,934
U	Science & Technology		12,581
V	Social Science		3,693
W	Public & Societal Benefit		62,936
X	Religion-Related		108,217
Y	Mutual & Membership Benefit		182,935
Z	Unknown or Unclassified		151,667

To further supplement this dataset, I gathered real Gross Domestic Product (GDP) data for each state. The Bureau of Economic Analysis (BEA), an agency under the United States

Department of Commerce, collects this data.[20] The BEA collects and verifies this data for each united state and territory. For this thesis, I use only real GDP, so after accounting for national inflation, for each state. All the data is in a .CSV file with yearly dollar values for each state from 1997 – 2024.

I subset this data to only the years 2009 – 2023 and convert the real GDP dollar value to a growth percentage. Since I have a GDP value for 2008, I have the complete range for my final dataset.

Figure 1 shows the overall state GDP growth rate trend in the final data set.

[20] Bea.gov. "BEA: Download ZIP Files," 2026. https://apps.bea.gov/regional/downloadzip.htm.

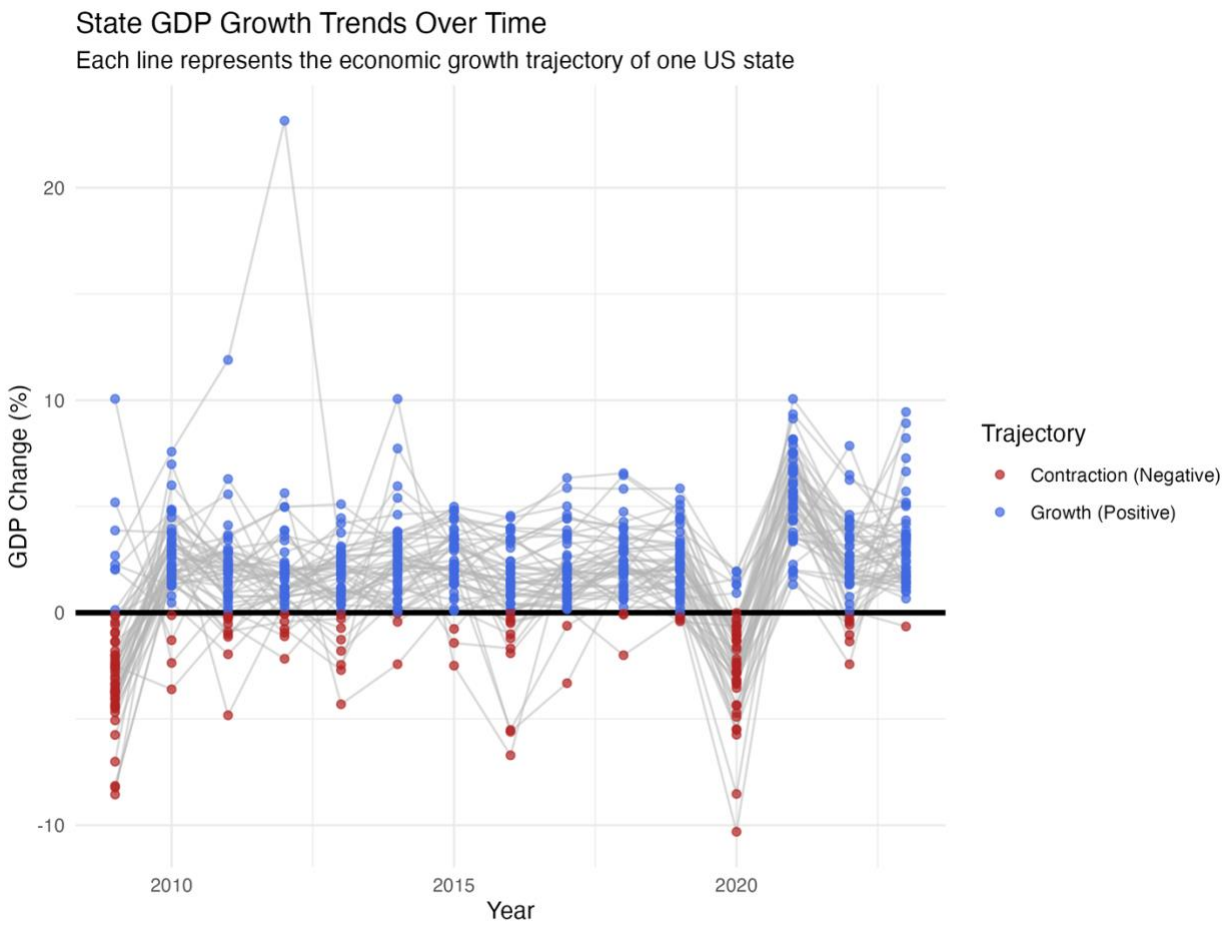

Figure 1. State GDP Growth, over time

These three data sources: Form 990 returns, Activity/NTEE code industries, and real GDP make up my final data set. No restrictions were placed on Activity/NTEE codes or real GDP data beyond subsetting the data to match my universe of non-profit observations. However, Form 990 returns include potential mistakes by filers, empty observations, and outliers so I must impose some data restrictions.

Restrictions

Firstly, according to 501(c)(3) status regulations, an organization that does not follow filing procedures for three consecutive years loses its tax-exempt status.[21] Additionally, an organization that is not contributing a public benefit is at risk of losings its special status. So, three consecutive years of zero program expenses, assuming this shows zero public benefit, is a reasonable restriction to make because these are organizations that may lose their non-profit status. This drops 1,470,587 observations.

Also, large institutions like hospitals and universities have significantly greater legal and financial guidance than smaller organizations and operate differently.[22] Shon and colleagues find that changing revenue concentrations - the primary experimental variable in their research - does not have a significant effect on hospital or school spending patterns. Further, these institutions represent some of the largest outliers in my dataset and demand for their services is likely inelastic. Mortensen and Chen demonstrate that hospital visit rates are the same before and during a recession, supporting inelastic demand.[23] For universities, demand for their services does increase during a recession like the other non-profit types, but so does total supply of the education, since enrollment increases accordingly.[24] Thus, universities are unique outliers amongst the overall trend of non-profit organizations and including them would skew the average behaviors for the rest of the Education industry. This drops 558,692 observations.

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[21] Irs.gov. "Automatic Revocation of Exemption | Internal Revenue Service," 2017. https://www.irs.gov/charities-non-profits/automatic-revocation-of-exemption.
[22] Shon, Jongmin, Madinah F. Hamidullah, and Lindsey M. McDougle. "Revenue structure and spending behavior in nonprofit organizations." The American Review of Public Administration 49, no. 6 (2019): 662-674.
[23] Mortensen, Karoline, and Jie Chen. "The Great Recession and Racial and Ethnic Disparities in Health Services Use." JAMA Internal Medicine 173, no. 4 (February 25, 2013): 315. https://doi.org/10.1001/jamainternmed.2013.1414.
[24] Schmidt, Erik P. "Postsecondary Enrollment before, during, and since the Great Recession. Population Characteristics. Current Population Reports. P20-580." US Census Bureau (2018).

Next, I must drop observations filing with negative values for financial variables. Shon and colleagues enforce a similar restriction in their research on non-profit financial health.[25] Though technically possible, this restriction makes the regressions easier to compute. It is also possible some organizations incorrectly file their annual reports, creating negative values for variables such as total revenue or expenses and those filing errors should be removed. This drops 85,230 observations.

Fourth, I restrict my dataset to only organizations with a headquarters in the United States. My economic variables are based in the fifty united states, and I do not have data for countries or states outside of this scope. Also, I hope for potential policy or business recommendations from the results to be relevant for the universe of organizations and governments involved in the dataset and should exclude international organizations to remove biasing these recommendations. This drops 47,049 observations.

Lastly, I remove singletons from the data, so that every regression now uses the same sample size. This is a necessary step for properly applying fixed effect variables, specifically on the firm/business level. For the regression to properly compute at the firm level, each organization must have at least two observations, so cannot be a singleton with one observation. This drops 63,501 observations.

From all these data restrictions, this limits the data set to 2,880,054 observations (from 5,002,756) across 391,529 unique organizations (from 738,790). Taking the log() distribution of key variables of this final dataset is now much more normally balanced, in contrast to before the restrictions or calculating log() values. Before, revenue and expense distributions were doubly

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[25] Shon et. al. "Revenue structure and spending behavior." (2019).

skewed, with strong right skewing from large organizations as well as a substantial number of singletons or null values concentrated at zero.

Description

Viewing the data with restrictions, the double skewing is no longer present because observing the log() observations successfully eliminate the effect of outliers. The data is mostly normally skewed, except with still a large concentration at zero. However, this is expectable and certainly plausible when considering non-profit operating procedures.

Firstly, many organizations experiencing financial difficulties may still be present in the dataset. The restriction is only for organizations that have no program expenses for three or more years, so this still includes organizations that may not earn a consistent revenue. Secondly, some organizations fund their operations via grants, which follow a cycle. An organization may receive a grant for a multi-year project, but must report all the revenue in year one, even if the grant is for two years of program expenses.[26]

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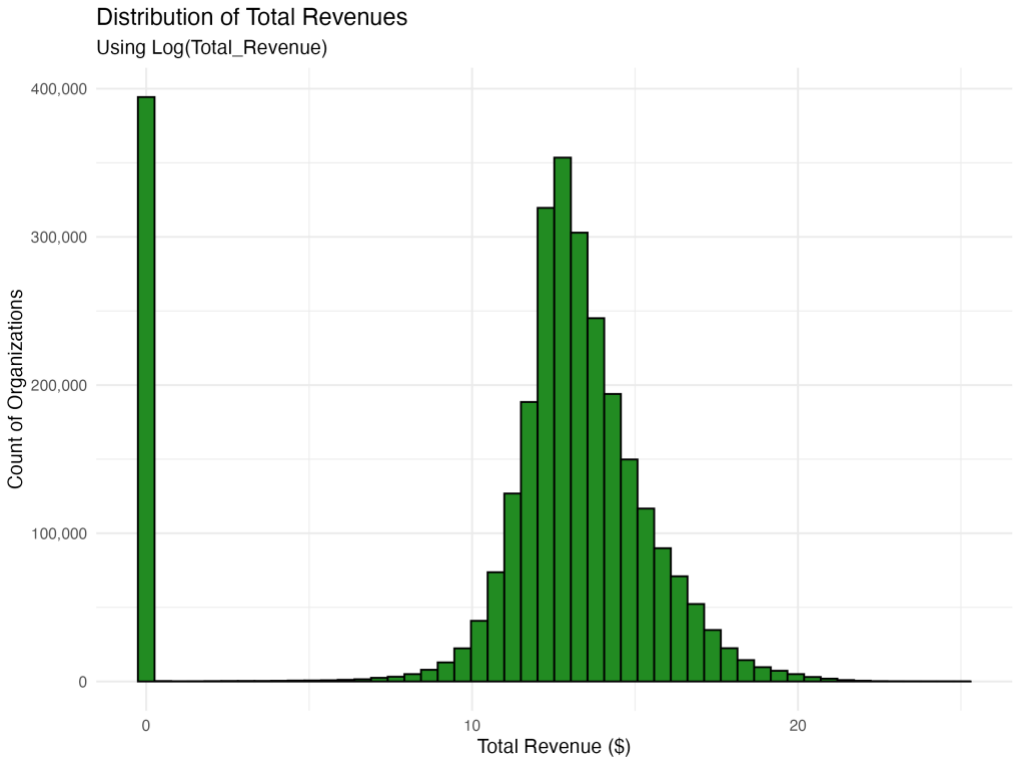
[26] Krebiehl, Nichole. "Altruic Advisors." Altruic Advisors Certified Public Accountants & Consultants, August 7, 2019. https://altruic.com/blog/revenue-recognition-for-nonprofit-grants#:~:text=In%20this%20case%2C%20since%20the,not%2Dfor%2Dprofit%20accounting..

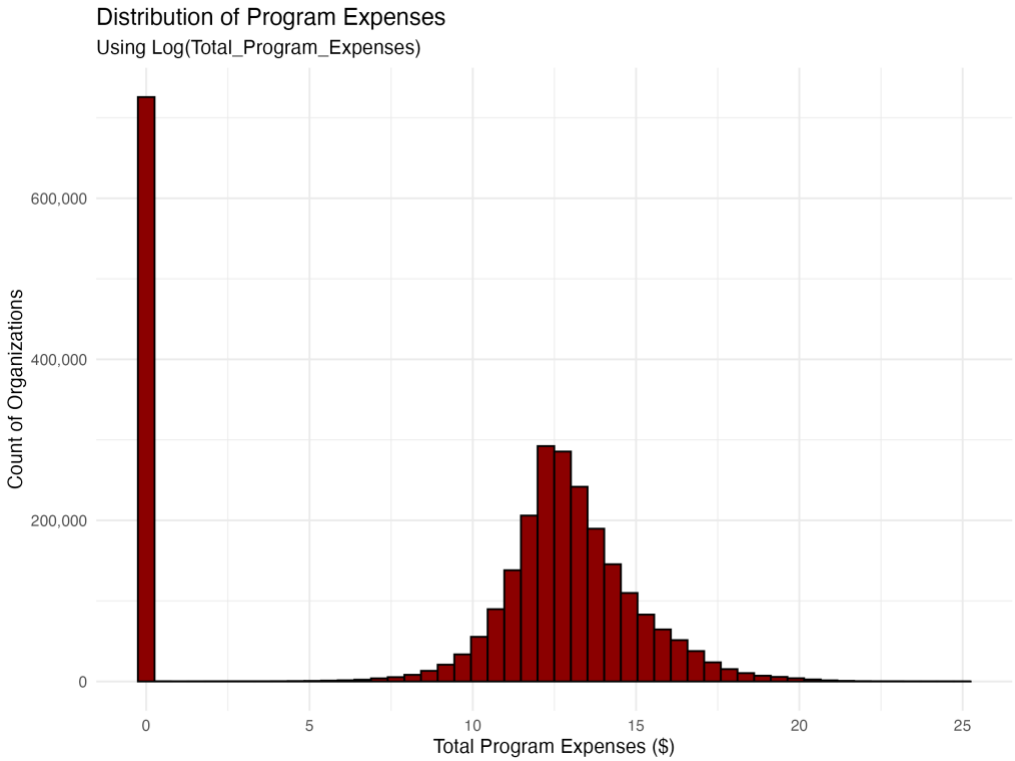
Figure 2. Distribution of Total Revenues



Now, looking at program expenses of the data set. Again, many financially unhealthy non-profits may be unable to fund their operations and report zero program expenses. Regardless, organizations that choose to or must cease program expenses, without dissolving the organization entirely, are extremely valuable for this thesis, which is about financial health and macroeconomics of non-profit organizations.

Another consideration, and potential bias for this thesis is non-profit organizations that operate without a budget. For example, organizations that rely on volunteers to mentor students, or business associations that connect people virtually. An organization with zero program expenses may or may not be an organization with zero social benefit however it is impossible to determine this from financial data alone. As such, I conservatively assume an organization with no program expenses is providing no social benefit.

Figure 3. Distribution of Program Expenses



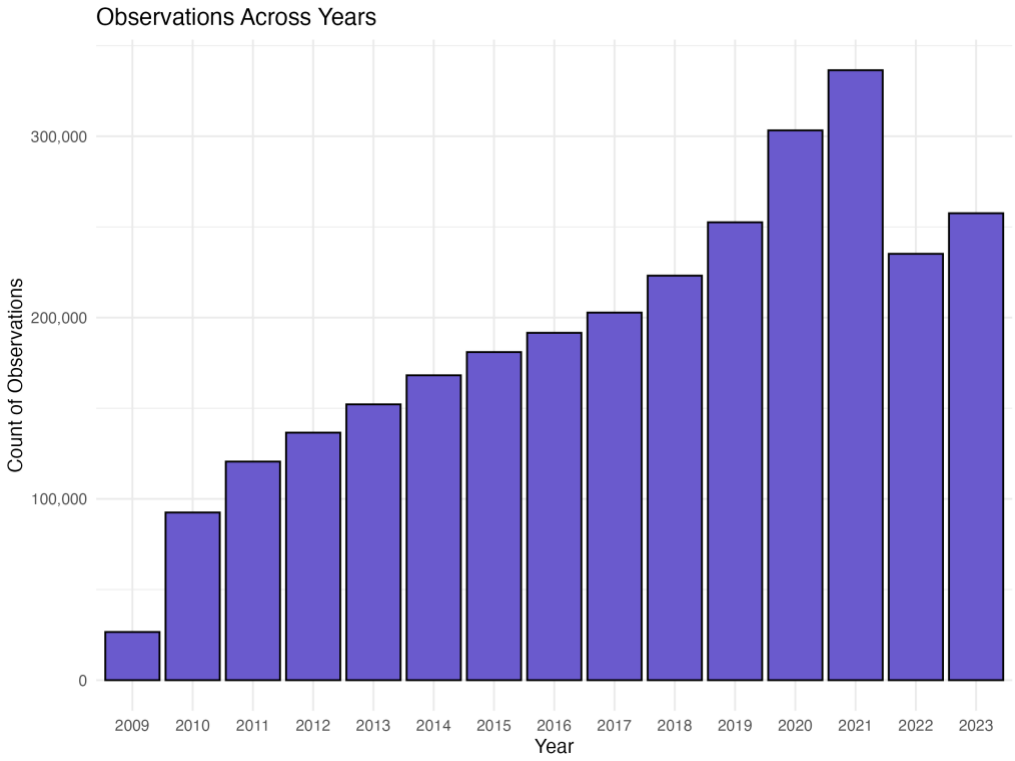
The graph below also shows the total number of non-profit organizations filing each year in the data set. The growth overtime in the number of organizations is expectable with rising wealth, country population, and a general rise of social entrepreneurship and non-profit funding following the recovery of the 2008 recession.[27] Jack and Patterson observe the large growth in organizations since 2009 is primarily concentrated in the Human Services (NTEE industry P), Environment (C), and Education (B) industries, accordingly.

Most notably, there is a steep drop-off of over 50,000 organizations from 2021 to 2022. It is impossible to know why each organization fell out of the dataset, but there are many trends that occurred. Firstly, the ending of Covid-relief funds from governments. Secondly, the IRS

[27] Salmon, Jack and Tanner, Patterson. "Growth of the Nonprofit Sector: More Charities, Better Solutions." Philanthropy Roundtable, March 31, 2025. https://www.philanthropyroundtable.org/resource/growth-of-the-nonprofit-sector-more-charities-better-solutions/

deadline was in 2022 requiring online tax filings. So, organizations could get confused and not file that year or following. Lastly, it could be the effect of the recession where revenue sources like donations dried up for non-profits and either shrunk to below the size requirement for Form 990 or dissolved entirely.

Figure 4. Number of Non-profits, by year



Across all variables, there is a wide range of observations. All financial variables have organizations reporting zero expenses or revenues, which is from young and small non-profits. On the other hand, some non-profits, such as those working for the federal government or in research and development sectors may be receiving and spending massive amounts of money compared to their peers.

Table 2. Descriptive Statistics for Selected Variables

Variable	Description	Observations	Mean	Standard Deviation	Minimum	Maximum
donation_revenue	Revenue from campaigns and direct donations	2,880,054	$9,065.60	356,498.20	$0.00	$127,014,579.00
fundraising_revenue	Revenue from fundraising events	2,880,054	$30,819.60	854,165.41	$0.00	$350,002,158.00
gdp_change_percent	Percentage change in real GDP, by state	2,880,054	2.26%	2.63	-10.31%	23.15%
government_revenue	Revenue from government grants	2,880,054	$609,078.65	15,940,399.11	$0.00	$11,069,708,938.00
investment_revenue	Net income from investments	2,880,054	$40,498.89	1,465,053.77	$0.00	$717,215,986.00
membership_revenue	Revenue from membership dues	2,880,054	$44,228.46	875,560.37	$0.00	$163,320,095.00
other_revenue	Miscellaneous revenue from other sources (i.e. sale of property, etc.)	2,880,054	$141,399.33	4,528,116.07	$0.00	$2,366,107,467.00
total_program_expenses	Total expenses directed towards programs. Represents the demand for the organization's services.	2,880,054	$6,636,765.48	150,719,949.06	$0.00	$71,256,037,114.00
total_revenue	Total revenue from all sources. Represents the supply of the	2,880,054	$9,090,684.18	174,916,735.29	$0.00	$75,101,306,911.00

Variable	Description	Observations	Mean	Standard Deviation	Minimum	Maximum
	organization's services.					
industry	Primary activity/industry category	26*				
organization_ein	Unique identifier (Employer Identification Number)	391,529*				
state	State where the organization is located	50*				
year	Tax year of the filing and GDP record	15*			2009	2023

* Observations for categorical variables represent unique counts.

Methodology

Independent Variables

My main explanatory variable is the percent change in each state's real gross domestic product (GDP) because it is a common metric to study economic conditions. This variable is adjusted for inflation, so fluctuations reflect real economic changes. I use state-level data rather than national GDP, so it captures the unique economic conditions that most impact the recipients and donors of a given non-profit.

Dependent Variables

Citizens have strong preferences for non-profits to expand services, especially in economically tough times.[28] For my research to best capture this preference through the data, I capture the change in program expenses year-to-year. So, my first key dependent variable is "Total Program Expenses" because it measures the total money spent on activities advancing each non-profits mission statement. Other research may use more direct survey data to measure 'direct impact' towards the mission in a recession,[29] however I do not have the opportunity to survey for such data and use program expenses as a proxy for impact. Thus, for the market of a given non-profit's goods/services, program expenses represent the demand side of this function.

My second key dependent variable is "Total Revenues" because it measures the total supply of cash and other incomes available to a given non-profit each year. While program expenses reflect the intent to meet recipients or economic needs, total revenues represent the supply side of this function. It is the actual financial capacity of the organization to sustain or

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[28] Exley, Lehr, and Terry. ""Nonprofits in good times and bad times."
[29] Altamimi, Hala, and Liu. "The nonprofit starvation cycle."

expand its programs. By accounting for revenues, I can determine the change in program expenses explainable by budget constraints rather than responses to economic condition.

Regression Models

For each model, I try to run it for the demand and supply side of the non-profit market.

Model 1

Starting with an ordinary least squares (OLS) model:[30]

$$\mathrm{Y}_i = \beta_0 + \beta_1 X_i + \varepsilon_i \tag{0}$$

Where Y_i is the dependent variable or outcome the model looks to predict, at the *i*-th observation. β_0 is the intercept of the equation when the effect is equal to 0. β_1 is the slope or magnitude of the effect of some explanatory variable X on dependent variable Y. X_i is the intendent or explanatory variable the model looks to analyze, at the *i*-th observation. ε_i is the error term and includes all effects that cannot be observed in the model.

Using the OLS model to compute a simple regression model can show the broadest correlation between the economy and program expenditure of all the organizations forms the model:

$$\log(Program\ Expenses) = \beta_0 + \beta_1(GDP\ Change\ \%) + \varepsilon \tag{1.1}$$

$$\log(Total\ Revenue) = \beta_0 + \beta_1(GDP\ Change\ \%) + \varepsilon \tag{1.2}$$

Where $\log(Program\ Expenses)$ and $\log(Total\ Revenue)$are my key dependent variables. Program expenses for a given organization and year are used to represent the total demand for an organization's philanthropic services, taken at the log level to diminish the effects of outliers. Total revenues for a given organization and year are used to represent the total supply

[30] DeHaan, Ed. "Using and interpreting fixed effects models." (2021).

of an organization's services, taken at the log level to diminish the effects of outliers. β_0 still represents the intercept of the equation. $\beta_1(GDP\ Change\ \%)$ is my explanatory variable for its effect on program expenses, change in macroeconomic conditions for a given year from the prior year, taken as a percentage to improve interpretation. ε still represents all unobserved effects.

This is useful because it analyzes the panel data and will show a baseline level of correlation if non-profits have a countercyclical trend. Drawing on concepts from stakeholder theory, organizations are altruistic and restructure their organization knowing times are worse for their constituents.[31] Or, like the rest of the economy, non-profit organizations must downsize in tough times and reduce expenses.

Interpreting the results of this model would show that for a one percentage point increase in real GDP, non-profit program expenses will increase/decrease by a percentage amount.

However, this model will suffer from omitted variable bias (OVB) and have imprecise results because of it. The results may be statistically significant since it is such a large dataset with many observations but will not have much explanatory power because non-profit organizations naturally will expand as the economy expands, or vice versa.

So, to improve the model there must be controls for variables that are constant across entities but still change over time and space. Such as yearly shocks, state policies, industry practices, and firm culture - to name a few potentially omitted variables.

Model 2

$$\log(Program\ Expenses)_{year} = \beta_0 + \beta_1(GDP\ Change\ \%) + \gamma_{year} + \varepsilon \qquad (2.1)$$

$$\log(Total\ Revenue)_{year} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year} + \gamma_{year} + \varepsilon \qquad (2.2)$$

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[31] McGlinch, James Thomas. "Powerful Stakeholders and the Management of Corporate Restructuring." University of Pennsylvania, 2023.

Where subscript $year$ is the variation of each year for every variable that includes yearly variation. Each year is a fixed constant in this model, thus becoming a 'fixed effect.' γ_{year} captures all other unobserved effects connected to each year.

Controlling for each year improves the model because it controls for any shocks that would affect all the organizations each year, so β_1 reflects a more precise relationship between the change in GDP and program expenses. It isolates variation to what is more plausibly tied to yearly GDP fluctuations, rather than a generic trend across all time.

However, this model still suffers from OVB because organizations across different states can face quite different economic environments. A non-profit operating in a highly regulated state may behave differently than one with a different set of conditions.[32]

To improve this model, I should add controls for differences across states to capture variations in state-level policy, culture, and economic conditions that affect how non-profits allocate their program expenses.

Model 3

Adding state fixed effects to the model controls for this variation across geography. The following model absorbs differences across states that are the same for organizations in the same state. This effect is represented by α_{state} and no longer biases β_1, a focus of this thesis, in the model.

$$\log(Program\ Expenses)_{year,state} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \alpha_{state} + \varepsilon \quad (3.1)$$

$$\log(Total\ Revenues)_{year,state} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \alpha_{state} + \varepsilon \quad (3.2)$$

Where subscript $state$ is the variation for each state isolated in the model, becoming a state fixed effect. α_{state} is all state-level variation of the result not explained by another variable.

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[32] Salamon, Lester M., and Stefan Toepler. "Center for Civil Society Studies." (2000).

Interpreting this model is like Models 1, 2, and the following fixed-effects models. The estimate for β_1 will be less biased than prior models and calculated as an average of the effect for each year and state.

However, many omitted variables may still exist in this model because of industry-wide trends and firm-specific practices.

To improve this model further, Mauri and Max (1998) find that it would be useful to control for the industry of companies and at the individual, firm level.[33] This can help refine the results because companies have similar strategies to competitors. Further, each company has unique set of resources, even in the same industry, creating a unique effect of firm financial choices for this thesis. Holding firms as a constant would absorb most if not all industry fixed effects, since each firm only belongs to one industry, but the comparison between the two models will be useful in comparison and heterogeneity tests.

Model 4

Adding industry fixed effects controls for unobserved, time-invariant characteristics that are unique to specific non-profit sectors creates the following demand and supply -side models.

$$\log(Program\ Expenses)_{year,state,industry} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \alpha_{state} + \tau_{industry} + \varepsilon \quad (4.1)$$

$$\log(Total\ Revenues)_{year,state,industry} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \alpha_{state} + \tau_{industry} + \varepsilon \quad (4.2)$$

Where subscript $industry$ is the variation for each industry isolated in the model, becoming an industry fixed effect. $\tau_{industry}$ captures all other unexplained variation across types of non-profits (e.g., healthcare organizations versus arts and culture organizations).

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[33] Mauri, Alfredo J., and Max P. Michaels. "Firm and industry effects within strategic management: An empirical examination." Strategic management journal 19, no. 3 (1998): 211-219.

This is useful because non-profits within the same industry likely share similar strategies, funding sources, and responses to shocks. By holding the industry constant, β_1 becomes even less biased, isolating the effect of GDP changes from industry-wide trends.

However, this model still faces omitted variable bias because even within the same state and industry, individual non-profit organizations have unique internal cultures, donor networks, reputations, and financial conditions. To improve this model and eliminate these biases, I must control for these firm-specific traits.

Model 5

Adding firm fixed effects acts as the final control for any time-invariant characteristics of a specific non-profit organization.

$$\log(Program\ Expenses)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (5.1)$$

$$\log(Total\ Revenues)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (5.2)$$

Where ω_{firm} is the firm fixed effect, capturing all unobserved, constant traits of the individual organization. Now, since a specific non-profit rarely changes its state headquarters location or industry classification, introducing firm fixed effects mathematically absorbs both the state (α_{state}) and industry ($\tau_{industry}$) fixed effects. So, ω_{firm} is highly effective to control for geographic, industry, and firm-specific variations simultaneously. For these reasons, this model acts as the definitive version of my supply-side regressions.

However, an issue remains on the demand-side. Available revenues fundamentally constrain an organization's ability to spend on programs. So, I should control for revenue each year. Else, a non-profit organization might appear to counter cyclically expand to fulfill increased demand when it is purely due to a large grant that must be used up.

Model 6

To control for revenue on the demand-side, I add $\beta_2 \log(Total\ Revenue)_{year}$ as a new explanatory variable to the equation to account for the budget of the organization. This becomes my final demand-side template model, which I will use for regression analysis:

$$\log(Program\ Expenses)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2 \log(Total\ Revenue)_{year} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (6)$$

Where $\beta_2 \log(Total\ Revenue)_{year}$ represents the total supply of cash and income available to the non-profit in a given year. Interpreting this model will be slightly different. Regressing $\log(x)$ on $\log(y)$ is interpreting the elasticity. So, a 1% increase in total revenue will be correlated with a β% change in program expenses.

This is useful because by holding the supply of funds constant, I can accurately observe shifts in the demand side and its effect on non-profit activities.

Model 7

Since Model 5 looked at total revenue as the supply side of organizations to observe trends, now I want to analyze deeper for each contribution of the supply. So, I want to repeat my supply-side analysis looking at specific sources of revenue for non-profits, which will show me which are affecting the trend on total revenue. Or which sources of revenue are cyclical vs countercyclical. Testing how GDP effects sources of revenue, this creates the specific regression models:

$$\log(Donations)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.1)$$

$$\log(Fundraising\ Events)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.2)$$

$$\log(Government\ Grants)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.3)$$

$$\log(Membership\ Fees)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.4)$$

$$\log(Investments)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.5)$$

$$\log(Miscellaneous)_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (7.6)$$

Where $\log(Donations)$, $\log(Fundraising\ Events)$, $\log(Government\ Grants)$, $\log(Membership\ Fees)$, $\log(Investments)$, and $\log(Miscellaneous)$ represent dependent variables, held fixed for each firm and year from the Form 990 returns data. Firm fixed effects absorb state and industry fixed effects, so they do not appear in the model.

This is useful because isolating the β_1 coefficient for each specific revenue source, I can better pinpoint which funding mechanisms might be most sensitive to macroeconomic fluctuations.

Model 8

Flipping Model 7, I want to use distinct revenue sources as explanatory variables to predict whether an organization can expand program expenses. Instead of measuring the effect of the economy on revenue sources, I want to measure the effect of revenue sources on reacting to economic conditions. Testing how GDP effects program expenses depending on the type of revenue, creates the following regression interaction models:

$$\log\ (\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Donations)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Donations)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.1)$$

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Fundraising\ Events)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Fundraising\ Events)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.2)$$

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Government\ Grants)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Government\ Grants)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.3)$$

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Membership\ Fees)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Membership\ Fees)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.4)$$

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Investments)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Investments)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.5)$$

$$\log(\text{Program Expenses})_{year,firm} = \beta_0 + \beta_1(GDP\ Change\ \%)_{year,state} + \beta_2(Miscellaneous)_{year,firm} + \beta_3(GDP\ Change\ \%)_{year,state} * \log(Miscellaneous)_{year,firm} + \gamma_{year} + \omega_{firm} + \varepsilon \quad (8.6)$$

Where $\beta_3(GDP\ Change\ \%) * (Revenue\ source)$ acts as an interaction term between the macroeconomic conditions and an organization's specific funding source.

This is useful because it considers the effect of macroeconomic condition types of revenue and an organization's program expenses but with further analysis. Using the interaction term can help determine the specific effect of non-profit revenue forms in its ability to change program expenses during economic conditions. Controlling for revenue forms isolates the effect

to only each form, which helps isolates the relationship between specific financial practices and countercyclicality.

Interpreting this model, a statistically significant, negative β_3 coefficient indicates a countercyclical driver. For example, as GDP decreases, an organization with a higher concentration of Government Grants will uniquely increase its program expenses.

However, this model still suffers from OVB from binding requirements for using grants and unmeasurable assumptions. For example, the model might show a strong countercyclical correlation with government grants because of legal mandates, instead of non-profit altruism.

Model 9

$$(\text{COVID Expansion})_{year,state,industry} = \beta_0 + \beta_1 \log(Donations)_{year,state,industry} + \beta_2 \log(Fundraising\ Events)_{year,state,industry} + \beta_3 \log(Government\ Grants)_{year,state,industry} + \beta_4 \log(Membership\ Fees)_{year,state,industry} + \beta_5 \log(Investments)_{year,state,industry} + \beta_6 \log(Miscellaneous)_{year,state,industry} + \gamma_{year} + \alpha_{state} + \tau_{industry} + \varepsilon \quad (9)$$

Where (COVID Expansion) is a Boolean value created for each organization to represent whether they spend more money on program expenses in year 2020 compared to 2019. Since each organization is either a true or false, I cannot use firm fixed effects because there is no variation in the dependent variable across the same firm. So, I include γ_{year}, α_{state}, and $\tau_{industry}$ fixed effects as control variables.

Interpreting this regression, my explanatory variables are financial variables taken at the log level, and my dependent variable is true or false. So, for a percentage point increase in a

specific revenue source, will be correlated with a percentage increase in the probability of expanding programs (COVID Expansion).

Results

First, I compare models 1-5 to see which has the greatest statistical significance and explanatory power. Each model adds new control variables and Table 3 shows those differences.

Table 3. Iterative Demand Side Results

	Model 1.1	**Model 2.1**	**Model 3.1**	**Model 4.1**	**Model 5.1**
(Intercept)	9.924***				
	(0.005)				
GDP Change %	-0.072***	-0.022***	-0.007**	-0.005*	0.006***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Num. Obs.	2880054	2880054	2880054	2880054	2880054
R2	0.001	0.041	0.043	0.154	0.847
FE: year		X	X	X	X
FE: state			X	X	X
FE: industry				X	X
FE: firm					X

* p < 0.05, ** p < 0.01, *** p < 0.001

Note: Regressing a non-profit's program expenses on state real GDP percentage change from prior year to current year, with each current year representing a Form-990 annual filing for tax-exempt status. Models 2.1, 3.1, 4.1, and 5.1 add fixed effects to the baseline model 1.1. The fixed effects are control variables for year, state, industry of non-profit, and firm's EIN number.

Model 1.1, the baseline demand-side results, representing a baseline Ordinary Least Squares (OLS) regression model, suggests a countercyclical relationship. For a one percentage point increase in real GDP, non-profit program expenses decrease by 7.2%. This result is significant at the 99% confidence level ($p < 0.01$).

However, as predicted, this model suffers from omitted variable bias (OVB). The coefficient (β_1) shrinks with each iterative model while the explanatory power (R^2) increases, and yet the statistical significance (p-value) remains high. Adding controls for year, state, and industry fixed effects reduces the average size of the coefficient while the observations and other variables remain the same. Model 4.1 shows for a one percentage point increase in real GDP, non-profit program expenses decrease by .5%. This model is still significant at the 95% confidence level ($p < 0.05$).

A pivotal shift occurs in Model 5.1, when adding firm fixed effects as the final degree of control in the model. The estimated effect becomes a positive sign, which indicates a procyclical trend. For a one percentage point increase in real GDP, non-profit program expenses increase by .6%, and this result is significant at the 99% confidence level ($p < 0.01$). With the jump in R^2 value, this indicates that individual firm traits and decisions explain most changes in program expenses. On average, when the economy grows and wealth increases, non-profit organizations receive some portion of it, equal to .6% according to the model, and spend it on program expenses. However, if the country's income falls, so do non-profits' incomes and they must downsize.

Looking at the supply side of the non-profit market and regression models in Table 4 shows a similar progression through the iterative models.

Table 4. Iterative Supply Side Results

	Model 1.2	**Model 2.2**	**Model 3.2**	**Model 4.2**	**Model 5.2**
(Intercept)	11.771***				
	(0.004)				
GDP Change %	-0.092***	-0.052***	-0.003	-0.003	0.006***
	(0.001)	(0.002)	(0.002)	(0.002)	(0.001)
Num. Obs.	2880054	2880054	2880054	2880054	2880054
R2	0.002	0.080	0.082	0.115	0.844
FE: year		X	X	X	X
FE: state			X	X	X
FE: industry				X	X
FE: firm					X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Regressing a non-profit's total revenue on state real GDP percentage change from prior year to current year, with each current year representing a Form-990 annual filing for tax-exempt status. Models 2.1, 3.1, 4.1, and 5.1 add fixed effects to the baseline model 1.1.The fixed effects are control variables for year, state, industry of non-profit, and firm's EIN number.

Starting at the baseline model, Model 1.2, again suggests a countercyclical relationship. For a one percentage point increase in real GDP, a non-profit's total revenues decrease by 9.2% and this result is significant at the 99% confidence level ($p < 0.01$). Although, this model is

without any controls and leaves a large amount of variance unexplained by the chosen explanatory variable.

By adding controls for year fixed effects in Model 2.2, state fixed effects in Model 3.2, and industry fixed effects in Model 4.2, more of the variance becomes statistically explained. At the same time, the size of the effect of the coefficient (β_1) decreases. Model 4.2, controlling for year, region, and industry, shows for a one percentage point increase in real GDP, a non-profit's total revenues decrease by .3% but this result is not statistically significant.

By holding firm specific traits as a constant in Model 5.2, the effect flips. For a one percentage point increase in real GDP, a non-profit's total revenue increases by 0.6%, and this result is statistically significant at the 99% confidence level ($p < 0.01$). The coefficient is a positive value, which tells us that non-profit revenue follows a procyclical trend. Revenues grow when the economy grows but shrink when the economy shrinks.

Table 5 shows the results for Model 6, which adds controls for individual firm choices by including total revenue as another variable for the demand side model. The table compares this revenue-controlled model with the firm-fixed effect models from above (Models 5.1 and 5.2).

Table 5. Non-profit Demand-side vs. Supply-side Cyclicality

	Model 5.1 (Demand Side)	**Model 5.2 (Supply Side)**	**Model 6 (Revenue Controlled)**
GDP Change %	0.006***	0.006***	0.001
	(0.001)	(0.001)	(0.001)
Log(Revenue)			0.823***
			(0.001)
Num. Obs.	2880054	2880054	2880054
R2	0.847	0.844	0.921
FE: year	X	X	X
FE: state	X	X	X
FE: industry	X	X	X
FE: firm	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Model 6 shows a different explanation for changes in a non-profit's program expenses; total revenue explains most of the variance. For a 1% increase in a non-profits total revenue, their program expenses increase by .82%, and this result is statistically significant at the 99% confidence level ($p < 0.01$). Meanwhile, the effect of GDP drops while total fit for the model, indicative in the R^2 value rising by 7.7%, improves. These results imply non-profit finances are elastic. A change in a non-profit's revenue changes their program offerings. If revenue declines, they must reduce expenses for their budget. If revenue remains constant, economic conditions may have only a slight cyclical trend effect, but this result is not statistically significant, and the standard error may bring the effect to zero.

Thus far, Models 5 and 6 shows that on average all non-profit organizations follow a cyclical trend with the economy that is constrained by their budget, or total available revenues. However, this result may be masking countercyclical behaviors that are invisible in the average, sum of all results, in the coefficient.

Table 6 shows the results of Model 6 for each quintile of total revenues.

Table 6. GDP Effect on Program Expenses, by Organization Revenue Size (Quintiles)

	Q1 (Smallest)	**Q2 (Small)**	**Q3 (Medium)**	**Q4 (Large)**	**Q5 (Largest)**
GDP Change %	0.009***	-0.002	0.001	0.002	-0.000
	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Log(Revenue)	0.861***	0.812***	0.809***	0.765***	0.696***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
Num. Obs.	372484	389014	639112	690965	788479
R2	0.946	0.906	0.834	0.861	0.919
FE: year	X	X	X	X	X
FE: state	X	X	X	X	X
FE: industry	X	X	X	X	X
FE: firm	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

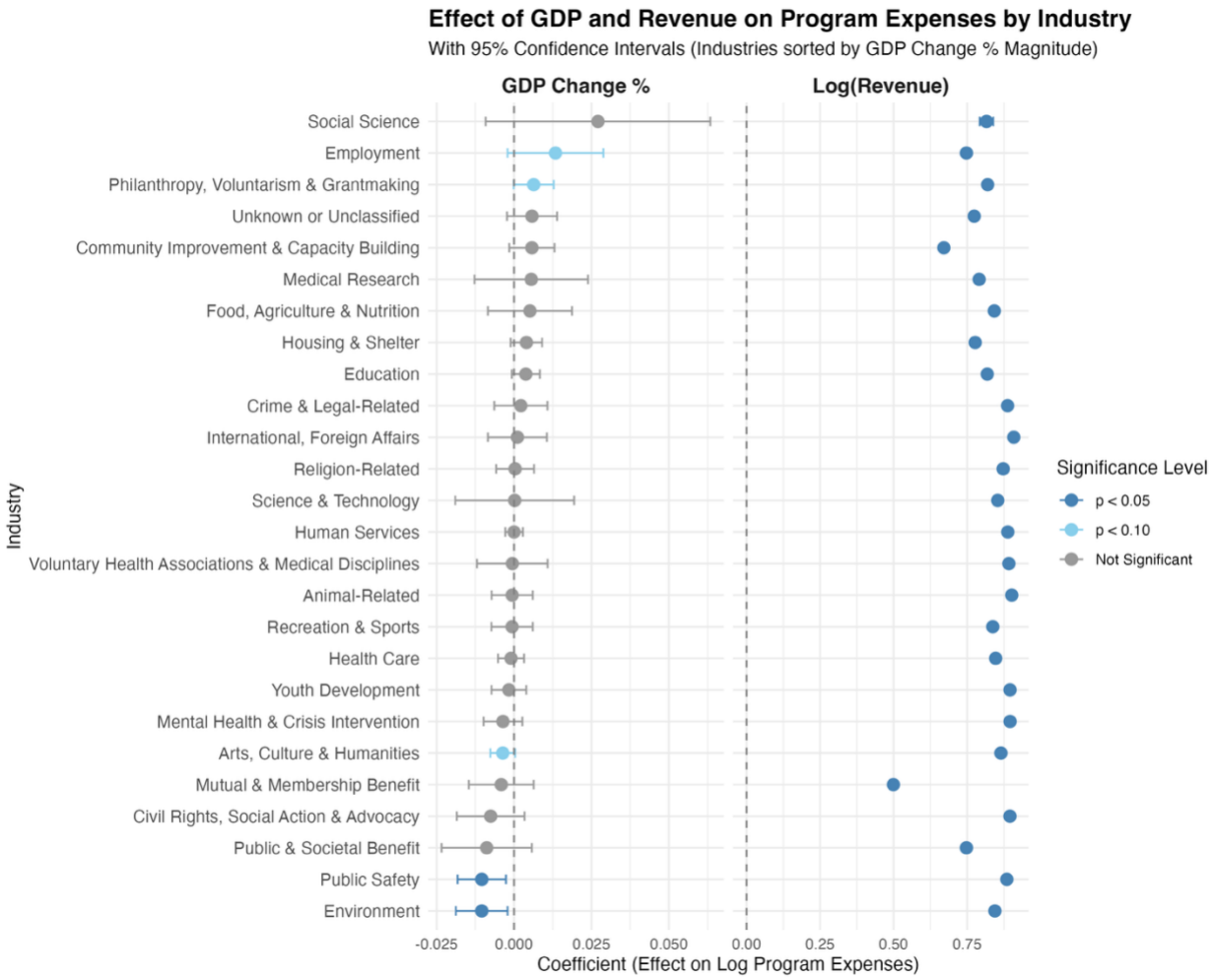
Note: Model 6, split into quintiles by revenue size according to the base year observation.

Starting at the results for the 20% smallest organizations, quintile 1 (Q1), shows a strong procyclical trend with economic conditions and available revenues. For a one percentage point increase in real GDP, non-profit program expenses increase by .9%. So, if the GDP decreases, so should program expenses.

Across larger organizations, quintiles (Q) 2-5, show a range in trend. Some may operate countercyclically while others stay procyclical. For a one percentage point increase in real GDP, non-profit's program expenses may decrease by .2% ranging to an increase of .2%, depending on revenue size. Further, across all sizes the budget constraint is evident. An increase in total revenue is always correlated with an increase in total program expenses; this result is statistically significant and a consistent effect across all organization sizes. As firms get larger, the effect of budget shrinks and the coefficient β_2 decrease from .861 to .696. Smaller firms appear to more budget elastic than larger ones and more effected by changes in economic conditions of their headquarter state.

Figure 5 (or Table 9 on page 55 in Appendix) summarizes the coefficients for Model 6 when split into each industry and tells a more detailed story than an average regression coefficient.

Figure 5. GDP Effect on Program Expenses by Industry



Across all organizations, there is a statistically significant effect of revenues in determining program expenses.

The GDP effect is only significant for some industries but shows a varying trend overall. Employment and Philanthropy Voluntarism & Grantmaking industries show a procyclical trend. For a one percentage point increase in GDP, program expenses increase by 1.3% and 0.6%, accordingly, and is significant at the 90% confidence level ($p < 0.1$). In the countercyclical direction, Public Safety and Environment industries have a statistically significant trend at 95% confidence level ($p < 0.05$). For a one percentage point increase in GDP, program expenses also decrease by 1%.

Most other industries show slight or almost zero coefficient with changing GDP. For example, Science & Technology and Human Services industries show exactly zero for coefficient B_1, though this result is not statistically significant. This low significance could be due to the inclusion of many outliers, especially close to zero dollars for many financial variables; or some industries have a low sample size n creating a high standard error relative to the coefficient.

Table 7 shows an altered supply side model, looking at how different revenue sources change due to economic conditions.

Table 7. Cyclicality of Non-Profit Revenue Sources

	Model 7.1 (Donations)	**Model 7.2 (Fundraising)**	**Model 7.3 (Govt Grants)**	**Model 7.4 (Membership)**	**Model 7.5 (Investment)**	**Model 7.6 (Other)**
GDP Change %	-0.000	0.004***	-0.001	0.002**	0.000	-0.003*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Num. Obs.	2880054	2880054	2880054	2880054	2880054	2880054
R2	0.754	0.705	0.780	0.859	0.877	0.712
FE: year	X	X	X	X	X	X
FE: state	X	X	X	X	X	X
FE: industry	X	X	X	X	X	X
FE: firm	X	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Model 6 and the heterogeneity tests show that program expenses are affected by available revenues, so Models 7.1-7.6 looks at how changes in GDP affect available revenues. The results show some revenue sources follow a procyclical trend, while some are countercyclical, and others have no clear trend. A one percentage point increase in GDP is correlated with a 0.4% increase in Fundraising and 0.2% increase in Membership revenue. These results are significant at 99% confidence level ($p < 0.01$). For the same increase in GDP, Other revenues decline by .3% on average and is at 99% confidence level ($p < 0.01$). Donations and Government Grants

revenue sources have no clear economic trend, both with a coefficient (β_1) near zero and no statistical significance.

Table 8 shows how each revenue source affected an organization's probability of expanding program expenses during the 2020 Covid-19 recession.

Table 8. Effect of Revenue Sources on Expanding During Covid

	Model 8 (Covid Expansion)	**Model 8 (Pre-Covid Robustness Check)**
Log(Donations)	0.003***	0.004***
	(0.000)	(0.000)
Log(Fundraising)	0.002***	0.005***
	(0.000)	(0.000)
Log(Govt Grants)	0.011***	0.008***
	(0.000)	(0.000)
Log(Membership)	-0.006***	-0.003***
	(0.000)	(0.000)
Log(Investment)	-0.005***	-0.003***
	(0.000)	(0.000)
Log(Other)	0.000*	-0.000***
	(0.000)	(0.000)
Num. Obs.	2318254	2171516
R2	0.079	0.058
R2 Adj.	0.079	0.058
FE: year	X	X
FE: state	X	X
FE: industry	X	X
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		

In these models, the dependent variable is a Boolean value, so the coefficients represent the change in the probability of expanding programs for every one percent increase in a specific revenue source.

Government Grants showed the strongest correlation. For a one percent increase in Government Grants revenue, there is a 0.01% increase in the probability of an organization expanding its programs and this result is highly significant at the 99% confidence level (p <

0.01). Donations and Fundraising revenue also show a positive and significant correlation with the probability of expanding programs at 0.003% and 0.002%, accordingly.

On the other hand, Membership and Investment revenues show a negative correlation with the probability of expansion. For a one percent increase in Membership Fees or Investment income, the probability of expanding program expenses decreases by 0.006% and 0.005%, accordingly. These results are also highly significant ($p < 0.01$).

These results are further affirmed by a pre-Covid robustness check of the model, which shows similar correlations across all the revenue sources.

Discussion

The iterative macro-models lend some support to altruistic theories about non-profit enterprise, that they expand their services during bad economic times to meet increased community need and scale back during booms, as argued by Pollifroni, Militaru, and Socaciu (2014).[34] However, the remaining the results of this thesis tell a different story.

The results of Model 5.1 and Model 5.2 align with recent research into non-profit macroeconomics by Exley, Lehr, and Terry (2023), who find that non-profit revenues across time, regions, and sectors follow a distinctly procyclical trend. Specifically, during economic downturns, revenues uniformly decline by about .7% for every 1% of income decline. The findings of Model 5, which utilizes firm-level fixed effects, validate this procyclical reality to a similar degree (revenues decline by about .6% for every 1% of income decline). Because this model controls for unobserved, time-invariant firm characteristics, it explains a large degree of variation in the dataset with high statistical significance. By isolating individual firm choices, it provides a highly accurate observation of how firms react to macroeconomic shifts.

The identical, positive coefficients found on both the supply (revenue) and demand (program expense) sides strengthen the conclusion that economic conditions and non-profit finances move in tandem. A change in state GDP has a direct effect on the market for non-profit goods and services, simultaneously impacting the non-profit's maximum supply of capital and its consumer-facing expenditures. This procyclical behavior observed in Model 6 supports DeVides and Harris (2025), who note that as total revenue increases, program expenses also increase. However, their research finds a different magnitude for this relationship, noting that for a 1%

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[34] Pollifroni, Militaru, and Socaciu. "Economic crises and the counter-cyclical profile of the Non-Profit Organisations."

increase in revenue, program expenses increase by 0.26%[35] versus .82% increase found in Model 6. This difference in magnitude is likely explained by their additional restrictions related to corporate governance and size, as well as my methodology of not splitting types of non-profit expenses in Model 6.

Clearly, organizations do not blindly adjust their spending precisely in tandem with revenue shocks. As demonstrated by Jones, Kitching, Roberts, and Smith (2012), charity managers are actively engaging in a "spend-save decision" where a marginal change in overall spending is substantially less than the marginal change in revenue. Instead of spending every marginal dollar, nonprofits intentionally smooth their spending across good times and bad times. When revenues increase by 1%, charities save approximately .31% of the change to increase liquid assets. Conversely, when revenues decline, charities mitigate deep cuts to program spending by drawing on these past savings.[36] These findings bring the results of Model 6 in line with the literature. This smoothing behavior, which is most pronounced in organizations with higher savings, shows financial management can moderate revenue constraints during economic shocks.

Repeating Model 6 across different industries provides further support for the budget constraint theory. While some sectors remain insulated from broader state GDP changes, they are still significantly affected by changes in their specific revenues. This reinforces the principle that an organization's operational capacity is ultimately bound by its available capital.

Looking at Model 7, these results can be distinguished by sources reliant on individual decisions (membership, donations, fundraising) vs external factors (investments, government

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[35] DeVides, Zhanel B., and Erica E. Harris. "Nonprofit Governance and Cost Behaviors." Journal of Accounting, Auditing & Finance (2025): 0148558X251405308.
[36] Jones et al. "The spend-save decision"

grants, other). With this perspective, revenue sources depending on a person's willingness to spend or donate (individual decision) are more procyclical, on average, than revenue sources tied to broader institutional decisions (external factors).

An unexpected finding is that direct donations exhibit no clear cyclical trend, which appears to contradict basic economic intuition and prior literature. Bridgeland et al. (2009) observed a distinct decline in donations during the 2008 recession, [37] and theoretically, as aggregate wealth grows, the average household's capacity to donate should follow suit. Furthermore, Rooney et al. (2020) documented a positive trend in overall donations from 2009 to 2020.[38]

This discrepancy in Model 7 may be capturing a different mechanism, though, as the variance in GDP in this thesis is at the state level. It is highly plausible that as a non-profit observes declining wealth within its domestic donor base, it compensates by increasing out-of-state fundraising efforts. If this inter-state marketing is successful, capital from wealthier states may flow in to subsidize the declining state, dampening the procyclical effect. Additionally, because a large share of charitable giving originates from high-net-worth individuals, an opportunity cost dynamic may be at play. During periods of GDP and market growth, the opportunity cost of donating their capital is higher, so donors may prefer to retain wealth for high-yield investments (e.g., stock market) rather than donating. These intuitions may explain the coefficient of zero in model 7.1, as it may be observing highly optimized behaviors of individuals or non-profit managers in response to changes in economic conditions.

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[37] Bridgeland et al. "The quiet crisis"
[38] Rooney, Patrick, Sasha Zarins, Jon Bergdoll, and Una Osili. "The impact of five different tax policy changes on household giving in the United States." In Nonprofit Policy Forum, vol. 11, no. 4, p. 20200040. De Gruyter, 2020.

Model 8 is some of the most interesting and policy relevant results. While models 1-7 were about overall financial behavior, model 8 compares these revenue practices to a real economic shock, the Covid-19 recession. Although the coefficients are extremely small, this is expectable for a large dataset with a wide range of non-profit types and sizes. Additionally, the Covid-19 recession had unique circumstances that might have made any non-profit expansion more unlikely, since there were mandatory shutdowns across the economy, social distancing requirements, and supply chain issues. But the very high statistical significance and positive results implies these results are still mathematically true correlations about ideal financial practices.

The results highlight a stark contrast between discretionary or market-tied revenues and philanthropic giving. Organizations that may be more reliant on membership fees (e.g., Employment industry) and investment income (e.g., Philanthropy, Voluntarism & Grantmaking) have a negative correlation with program expansion (-0.006% and -0.005%, respectively). Economically, this aligns with consumer behavior during a recession: as unemployment spikes, households eliminate discretionary spending, such as club memberships or business association dues, thus starving these specific non-profits of operating cash. Simultaneously, a recession decreases investment yields, forcing organizations with large endowments to reduce or eliminating spending to preserve their capital. This is supported with the findings from Bridgeland et al., where disposable non-profit spending and foundation philanthropy declined in the Great Recession.[39]

However, the effect of direct donations and fundraising proved uniquely resilient, correlated with positively driving expansion (+0.003% and +0.002% probability per 1% of

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[39] Bridgeland et al. "The quiet crisis"

growth in prior years), and contrasting the findings from the Great Recession. Unlike restricted funds or illiquid assets, donations often represent agile, unrestricted cash flow. This is likely due to the liquidity factor of these revenue sources. Direct donations and fundraising yield unrestricted, liquid cash which can make an organization better able to adapt to the unique circumstances of Covid-19, such as transitioning to virtually provided services, changing work from home policies, and buying personal protective equipment. This liquidity can allow funding expansions immediately, rather than waiting for structural changes or foundation grant policies to respond.

The most effective driver of countercyclical expansion in Model 8 is rooted in public policy rather than individual giving. Revenues from government grants had the strongest positive correlation with program expansion (+0.01% probability per 1% of growth in prior years). In 2020, massive federal subsidies, specifically the CARES Act, injected unprecedented cash stimulus into the non-profit sector.[40] The results of Model 8 suggest that non-profits with pre-existing government grant reliance and connections were uniquely positioned to benefit from this stimulus. Consequently, these organizations were more likely to fulfill the theoretical altruistic, countercyclical theories proposed by Pollifroni, Militaru, and Socaciu (2014).[41] Public policy likely subsidized their budget constraints, improving the ability to respond to the increased demand of Covid-19 for non-profits goods and services.

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[40] Havens, Jason E, and Kelly L Hellmuth. "Nonprofits and the CARES Act: A Closer Look | Insights | Holland & Knight." Hklaw.com, 2020. https://www.hklaw.com/en/insights/publications/2020/03/nonprofits-and-the-cares-act-a-closer-look.

[41] Pollifroni, Militaru, and Socaciu. "Economic crises and the counter-cyclical profile of the Non-Profit Organisations"

Conclusion

This research sought to explore what specific financial practices non-profit organizations can follow during periods of economic growth that can correlate to capacity to expand during an economic downturn, when a non-profit's services are more needed. To answer this question, I ran longitudinal panel data regression analyses on non-profit tax returns, which offered detailed financial insights of an organization's activities. These regressions quantified the correlation between macroeconomic conditions (State GDP) and the non-profit's financial behavior across three distinct dimensions: the demand-side of the non-profit market and the effect of macroeconomic conditions in how organizations change their philanthropic programs offered; the supply-side of a non-profit's activities and the effect of macroeconomic conditions on revenue sourcing for an organization; and a revenue-controlled dimension and whether an organization responds to increased demand for philanthropy even under a budget constraint.

By analyzing this comprehensive dataset of U.S. non-profits from 2009 to 2023, this thesis confirms prior literature that the non-profit sector is largely procyclical. An organization's ability to expand its program expenses is constrained by its available revenue, which is constrained by economic conditions. Meaning that as the broader economy contracts, non-profit supply shrinks just as public demand for philanthropy peaks, like during a recession.

However, countercyclical behavior is still possible and observable. The data suggests that organizations can actively engage in financial smoothing, mitigating the severity of revenue shocks to protect program expenditures. Furthermore, the analysis of the Covid-19 recession reveals that improving the chances for expansion, such as to match expanded demand for food services, in a recession is possible, but it is highly dependent on an organization's revenue composition.

Future research should have an even more granular analysis of financial decision making. This thesis was limited to use summarized financial information, instead of analyzing the experiences of non-profit mangers who might prioritize and distribute different sources of revenue, monitor organization-specific financial risks, and rely on other stakeholders to influence program expenditure decisions. Thus, defining specific financial 'practices,' either from interviews to gauge industry 'recommended practices' or testing specific techniques from literature may be a more effective analysis.

Another limitation of this research was not looking into changes in the balance sheet and other accounting decisions. Future research should aim to dissect these mechanics further by examining detailed changes in organizational balance sheets, such as the specific reallocation of assets and liabilities during a shock.

Also, it might be useful to interview non-profit directors, to see if these considerations for withstanding recessions, operating countercyclically, and responding to increased demand of philanthropy during a recession are even active concerns by the leadership. This would vastly strengthen any findings regarding 'best' practices to follow.

Finally, pairing quantitative panel data with qualitative interviews of non-profit directors would provide invaluable context. This could help determine the extent to which countercyclical planning is even an active concern among leadership.

In conclusion, further work should be done to investigate these financial practices to help non-profit enterprise to develop more refined, evidence-based strategies to ensure that non-profits remain resilient precisely when society needs them most.

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Appendix

Appendix 1:

Table 9. Regression results for GDP Effect on Program Expenses, by Industry

	GDP Change %	Log(Revenue)
Recreation & Sports	-0.001	0.837***
	(0.003)	(0.002)
Public & Societal Benefit	-0.009	0.747***
	(0.007)	(0.005)
Mutual & Membership Benefit	-0.004	0.499***
	(0.005)	(0.005)
Community Improvement & Capacity Building	0.006	0.671***
	(0.004)	(0.003)
Arts, Culture & Humanities	-0.004+	0.865***
	(0.002)	(0.001)
Unknown or Unclassified	0.006	0.774***
	(0.004)	(0.003)
Environment	-0.010*	0.844***
	(0.004)	(0.003)
International, Foreign Affairs	0.001	0.908***
	(0.005)	(0.002)
Health Care	-0.001	0.847***
	(0.002)	(0.002)
Education	0.004	0.818***
	(0.002)	(0.001)
Food, Agriculture & Nutrition	0.005	0.842***
	(0.007)	(0.005)
Philanthropy, Voluntarism & Grantmaking	0.006+	0.820***
	(0.003)	(0.002)
Human Services	-0.000	0.888***
	(0.001)	(0.001)
Science & Technology	0.000	0.854***
	(0.010)	(0.006)
Mental Health & Crisis Intervention	-0.004	0.896***
	(0.003)	(0.002)
Religion-Related	0.000	0.872***
	(0.003)	(0.002)
Medical Research	0.006	0.791***
	(0.009)	(0.006)
Youth Development	-0.002	0.896***
	(0.003)	(0.002)
Housing & Shelter	0.004	0.777***
	(0.003)	(0.002)
Animal-Related	-0.001	0.901***
	(0.003)	(0.002)
Voluntary Health Associations & Medical Disciplines	-0.001	0.892***
	(0.006)	(0.003)
Crime & Legal-Related	0.002	0.887***

	(0.004)	(0.003)
Employment	0.013+	0.747***
	(0.008)	(0.006)
Civil Rights, Social Action & Advocacy	-0.008	0.895***
	(0.006)	(0.003)
Public Safety	-0.010**	0.885***
	(0.004)	(0.003)
Social Science	0.027	0.816***
	(0.019)	(0.012)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Appendix 2:

Link to code repository: https://github.com/jonathantamen/economics-honors-thesis